1. Spatial dependency and its evaluation. Let observed spatial regions be numbered $i = 1, 2, \ldots, n$. The adjacency matrix $\widetilde{W}$ is defined as an $n \times n$ matrix, whose (i, j)-th element equals one if region i is adjacent to region j, and zero otherwise. In setting up the regression model, we row-normalize $\widetilde{W}$ so as to normalize the effect of the number of neighboring regions, and denote the resulting matrix as $W : n \times n$. The W is also called the spatial weight matrix Elhorst (2014).

1.1. *Measures of spatial dependency of spatial data.* It is known that testing spatially-dependent data as independent will excessively detect significant differences. Therefore, when analyzing spatial data, it is necessary to know before analyzing whether the data are spatially-dependent or not.

Moran (1950) proposed I statistic for evaluating the presence of spatial dependency based on the auto-correlation of spatial data $y_1, \ldots, y_n$. Using a row-normalized matrix W, we obtain the following statistics: $I = \frac{\sum_i \sum_j w_{ij}(y_i - \overline{y})(y_j - \overline{y})}{\sum_i (y_i - \overline{y})^2}$. If the data are independent, I is close to 0, otherwise far from 0. Approximating the variance of I under the null hypothesis, a significance test is done using the standard normal distribution.

The spatial weight matrix W plays a crucial role in representing spatial proximity and distance relationships among regions. Ward and Gleditsch (2019) evaluated alternative definitions of spatial distance, including contiguity-based adjacency between regions, geodesic distance between national capitals, and adjacency among U.S. states defined by either the four cardinal directions or the eight-directional (queen-type) neighborhood structure. These alternative specifications of W allow for flexible modeling of spatial dependence consistent with geographical or network connectivity.

2. Spatio-temporal Regression Models Incorporating Spatial Dependence. We will review traditional spatial regression models in the univariate case. Let $\boldsymbol{y} = (y_1, \ldots, y_n)'$ be the response vector observed n regions, $X^* = \left[\boldsymbol{x}_1^*; \ldots; \boldsymbol{x}_n^*\right]' : n \times p$ be the design matrix, and $W : n \times n$ be a row-normalized adjacency matrix. Then, WX^* can be regarded as an additional design matrix (set of regressors) constructed from the neighborhood averages of the explanatory variables. In most case, the explanatory vector $\boldsymbol{x} : n \times 1$ In what follows, we may use the design matrix $X = X^* : n \times p$ or an augmented design matrix $X = \left[\, X^* \,;\, WX^* \,\right] : n \times 2p$. In many cases, for a regressor vector $\boldsymbol{x} : n \times 1$, its spatial lag $W\mathbf{x}$ tends to be a weak explanatory variable relative to $\mathbf{x}$. Therefore, when using an augmented design matrix, variable selection is essential.

2.1. *Definition of Spatial Regression Models.* Let an error vector of the regression model $\boldsymbol{\epsilon} : n \times 1$ follows a multivariate Gaussian $N(\mathbf{0}, \sigma^2 I)$. Spatial regression models extend the standard regression framework by incorporating the following models that explicitly account for spatial dependence (Elhorst, 2014).

$$\text{Spatial AutoRegressive Model (SAR)}: \quad \boldsymbol{y} = \rho W \boldsymbol{y} + X\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{1}$$

$$\text{Spatial Error Model (SEM)}: \quad \boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\xi}, \quad \boldsymbol{\xi} = \lambda W \boldsymbol{\xi} + \boldsymbol{\epsilon}, \tag{2}$$

$$\text{General Nesting Spatial model (GNS)}: \quad \boldsymbol{y} = \rho W \boldsymbol{y} + X\boldsymbol{\beta} + \boldsymbol{\xi}, \quad \boldsymbol{\xi} = \lambda W \boldsymbol{\xi} + \boldsymbol{\epsilon} \tag{3}$$

Note 1: These models are hierarchically nested: when $\lambda = \rho = 0$ in (3), the specification reduces to the ordinally regression model; when $\lambda = 0$, it reduces to (1); and when $\rho = 0$, it reduces to (2).

Note 2: In models (1) and (3), the expected vector $E[\boldsymbol{y}]$ is given by $(I-\rho W)^{-1}X\boldsymbol{\beta}$. Hence, spatial correlation transmits change of explanatory vector of region i to the expected responses of other regions. This is referred to as the **spillover effect** (Rüttnauer, 2025).
Note 3: To select an appropriate spatio-temporal regression model, one must jointly optimize over candidate probability specifications and p explanatory variables. Fitorianto et al. (2022) proposed VSSR (Variable Selection of Spatial Regression) approach, in which the optimal subset of explanatory variables is first identified by minimizing AIC under fixed spatial-parameter values for each probability model. A grid search is then performed over the spatial parameters, yielding the optimal model configuration that achieves the optimal model.

3. Proposal of Multivariate Spatio-Temporal Regression (MSTR) Models. Up to the previous section, we considered spatio-temporal regression models for a single response variable. Here, we model mutually related spatio-temporal data such as GDP and CO_2 observed over small regions. In what follows, we consider K series of response variables observed over n regions at time t. We propose a multivariate version of the GNS model and describe its parameter estimation based on the maximum likelihood method.

3.1. *Multivariate General Nesting Spatial (MGNS) models within the multivariate spatio-temporal regression.* Consider MSTR with K response vectors $\boldsymbol{y}_{1,t}$, ... , $\boldsymbol{y}_{K,t}$ observed at n regions and time t. Put the following vector and matrices by

$$\boldsymbol{y}_t=\begin{bmatrix}\boldsymbol{y}_{1,t}\\ \vdots\\ \boldsymbol{y}_{K,t}\end{bmatrix},\ R=\begin{bmatrix}\rho_{11}&\cdots&\rho_{1K}\\ \vdots&\ddots&\vdots\\ \rho_{K1}&\cdots&\rho_{KK}\end{bmatrix},\ A=\begin{bmatrix}\alpha_{11}&\cdots&\alpha_{1K}\\ \vdots&\ddots&\vdots\\ \alpha_{K1}&\cdots&\alpha_{KK}\end{bmatrix},\ \Lambda=\begin{bmatrix}\lambda_{11}&\cdots&\lambda_{1K}\\ \vdots&\ddots&\vdots\\ \lambda_{K1}&\cdots&\lambda_{KK}\end{bmatrix} \tag{4}$$

where R, A, and Λ respectively are unknown coefficient matrices giving spatial lag of responses, auto-regressive terms, and spatial lag of errors. Then, the MGNS model within the MSTR framework is defined by

$$\boldsymbol{y}_t=(R\otimes W)\,\boldsymbol{y}_t+X_t^*\boldsymbol{\beta}^*+(A\otimes I_n)\,\boldsymbol{y}_{t-1}+\boldsymbol{\xi}_t \tag{5}$$

where $W:n\times n$ is the row-normalized adjacency matrix, $X_t^*=\begin{bmatrix}X_{1,t}^*&\cdots&\mathbf{O}\\ \vdots&\ddots&\vdots\\ \mathbf{O}&\cdots&X_{K,t}^*\end{bmatrix}$ is a design matrix with $X_{tk}^*:n\times p_k^*$ $(k=1,...,K)$, $\boldsymbol{\beta}^*:p^*\times 1$ is a regression coefficient vector, and $p^*=p_1^*+\cdots+p_K^*$. Further, we assume the error vector $\boldsymbol{\xi}_t=\left(\boldsymbol{\xi}_{1,t}',...,\boldsymbol{\xi}_{K,t}'\right)'$ is distributed as spatially-dependent Gaussian as

$$\boldsymbol{\xi}_t=(\Lambda\otimes W)\,\boldsymbol{\xi}_t+\boldsymbol{\epsilon}_t,\ \ \boldsymbol{\epsilon}_t\sim N({}_{Kn},\ \Sigma\otimes I_n) \tag{6}$$

where $\Sigma=\left(\sigma_{k,\ell}\right):K\times K$ is an unknown covariance matrix. Thus, the first, second and third terms of the right-hand side of Eqn (5) give spatial lag Y, exogenous effects and AR(1). The model given by Eqn (5) is a direct generalization of the GNS with single response. We note here that Although the design matrices $X_{1,t}^*,...,X_{K,t}^*$ may contain some explanatory variables in common, the regression coefficients are expected to differ across response variables; therefore, they are specified as separate matrices.

Here, we introduce of **the regular form** of MSTR framework. Let the observed response vectors at $t-1$ as $Y_{t-1}=\left[\ \boldsymbol{y}_{1,t-1}\,;\ldots;\boldsymbol{y}_{K,t-1}\ \right]:n\times K$, and AR(1) coefficients $\boldsymbol{\alpha}_k=(\alpha_{k1},...,\alpha_{kK})'$ $(k=1,...,K)$. Then, the second and third terms of the right-hand side of Eqn

(5) is simply combined as

(7)
$$\begin{bmatrix} X^*_{1,t} \; Y_{t-1} \cdots & \mathbf{O} & \mathbf{O} \\ \vdots \quad \ddots \quad \ddots & \vdots & \vdots \\ \vdots \quad \vdots \quad \ddots & \ddots & \vdots \\ \mathbf{O} \quad \mathbf{O} & \cdots & X^*_{K,t} \; Y_{t-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}^*_1 \\ \boldsymbol{\alpha}_1 \\ \vdots \\ \boldsymbol{\beta}^*_K \\ \boldsymbol{\alpha}_K \end{bmatrix} = \begin{bmatrix} X_{1,t} & \cdots & \mathbf{O} \\ \vdots & \ddots & \vdots \\ \mathbf{O} & \cdots & X_{K,t} \end{bmatrix} \boldsymbol{\beta} \; = \; X_t \, \boldsymbol{\beta}$$

where $X_{k,t} = [X^*_{k,t}\,;\, Y_{k,t-1}] : n \times p_k \;(p_k := p^*_k + K^2)$ are composite design matrices with $\boldsymbol{\beta}_k = (\boldsymbol{\beta}^{*'}_k,\, \boldsymbol{\alpha}'_k)' : p_k \times 1$ and $\boldsymbol{\beta} := (\boldsymbol{\beta}'_1, ..., \boldsymbol{\beta}'_K)' : p \times 1$ is a combined coefficient vector with $p := p_1 + \cdots + p_K$. Then, Eqns (5) – (7) are unified to **the standard form of MGNS models** as

(8) $$\boldsymbol{y}_t = (R \otimes W)\boldsymbol{y}_t + X_t\,\boldsymbol{\beta} + (I_{Kn} - \Lambda \otimes W)^{-1}\boldsymbol{\epsilon}_t \quad \text{with} \quad \boldsymbol{\epsilon}_t \;\sim\; N(\mathbf{0}_{Kn},\; \Sigma \otimes I_n)$$

Note 5: The expected vector of $\boldsymbol{y}_t$ is given by $E[\,\boldsymbol{y}_t\,] = (I_{Kn} - R \otimes W)^{-1}\, X_t\, \boldsymbol{\beta}$. Hence, the proposed model is able to explain **the spill-over effect**.

3.2. *Parameter estimation by the maximum likelihood principle.* The log likelihood of $\boldsymbol{y}_t$ is given by

(9)
$$\ell(R, \Lambda, \boldsymbol{\beta}, \Sigma) = -\tfrac{Kn}{2}\log(2\pi) + \log\left|I - R \otimes W\right| + \log\left|I - \Lambda \otimes W\right| - \tfrac{n}{2}\log|\Sigma| - \tfrac{1}{2}Q$$

where

(10)
$$Q = \boldsymbol{z}'\,(\Sigma^{-1} \otimes I_n)\,\boldsymbol{z} \;\text{ with the residual vector: }\; \boldsymbol{z} = (I - \Lambda \otimes W)\big\{\boldsymbol{y}_t - (R \otimes W)\boldsymbol{y}_t - X_t\,\boldsymbol{\beta}\big\}.$$

If R and Λ are both known matrices, MGNS model is just a multivariate regression model with correlated errors. Recall the matrix derivative identity : $\partial \log |\Sigma^{-1}| / \partial\, \Sigma^{-1} = \Sigma$, likelihood equations with respect to $\boldsymbol{\beta}$ and $\Sigma^{-1} = [\,\sigma^{k,\ell}\,]$ are given by

(11) $$\partial\,\ell / \,\partial \boldsymbol{\beta} = -\tfrac{1}{2}\,\partial Q / \,\partial \boldsymbol{\beta} = -\tfrac{1}{2} \times (-2) \times X'_t\,(I_n - \Lambda \otimes W)'\,(\Sigma^{-1} \otimes I_n)\,\boldsymbol{z} = \mathbf{0}$$

(12) $$\partial\,\ell / \,\partial \sigma^{k,\ell} = \frac{n}{2}\sigma_{k,\ell} - \tfrac{1}{2}\,\boldsymbol{z}'\left\{\mathbf{e}_k \mathbf{e}'_\ell \otimes I_n\right\}\boldsymbol{z} \;=\; \frac{n}{2}\sigma_{k,\ell} - \frac{n}{2}\boldsymbol{z}'_k \boldsymbol{z}_\ell = 0$$

where $\mathbf{e}_k = (\,\underbrace{0, \ldots, 0}_{k-1},\; 1\,, \underbrace{0, \ldots, 0}_{K-k}\,)' : K \times 1$. Thus, we have

(13)
$$\widehat{\boldsymbol{\beta}} = \widehat{\boldsymbol{\beta}}(\Lambda, \Sigma) \;=\; B\,\boldsymbol{y}, \;\; \text{where}$$

(14)
$$B = \left\{X'_t\,(I - \Lambda \otimes W)'\,(\Sigma^{-1} \otimes I)\,(I - \Lambda \otimes W)X_t\,\right\}^{-1} X'_t\,(I - \Lambda \otimes W)'\,\left(\Sigma^{-1} \otimes I\right)(I - \Lambda \otimes W)$$

(15)
$$\widehat{\Sigma} = \widehat{\Sigma}(R, \Lambda, \boldsymbol{\beta}) \;=\; \frac{1}{n}\begin{pmatrix} \boldsymbol{z}'_1\,\boldsymbol{z}_1 & \cdots & \boldsymbol{z}'_1\,\boldsymbol{z}_K \\ \vdots & \ddots & \vdots \\ \boldsymbol{z}'_K\,\boldsymbol{z}_1 & \cdots & \boldsymbol{z}'_K\,\boldsymbol{z}_K \end{pmatrix}$$

where $\boldsymbol{z}_k \;(k = 1, ..., K)$ are n-dimensional sub vectors of the residual vector $\boldsymbol{z} = [\,\boldsymbol{z}'_1, ..., \,\boldsymbol{z}'_K\,]'$, which was defined by (10). Here, the variance-covariance matrix of $\widehat{\boldsymbol{\beta}}$ is obtained by the following matrix Ψ as

(16) $$\Psi = B\;\mathrm{Var}(\boldsymbol{y})\,B' = \left\{X'_t(I - \Lambda \otimes W)'\,\left(\Sigma^{-1} \otimes I\right)(I - \Lambda \otimes W)X_t\right\}^{-1}$$

which will be used for significance level evaluation.

3.3. *Estimation procedure of GNS within MSTR framework.* Let partition the design matrix as $X_t = \left[\ X'_{1,t}\,;\ \dots\ ;\ X'_{K,t}\ \right]'$ with $X_{k,t} : n \times p$. Then, the estimation procedure will be as follows.

S1 Estimate initial parameter by each GNS.

a) For each $k = 1, 2, ..., K$, estimate λ_{kk} and σ_{kk} of the GNS given by

$$\boldsymbol{y}_{k,t} = \rho_{kk} W \boldsymbol{y}_{k,t} + X_{k,t}\boldsymbol{\beta} + (I - \lambda_{kk} W)^{-1}\boldsymbol{\epsilon}_{k,t} \ \text{ with } \ \boldsymbol{\epsilon}_{kt} \sim N(\mathbf{0}_n, \sigma_{kk} I) \tag{17}$$

b) Set $R = \begin{pmatrix} \widehat{\rho}_{11} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \widehat{\rho}_{KK} \end{pmatrix}$, $\Lambda = \begin{pmatrix} \widehat{\lambda}_{11} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \widehat{\lambda}_{KK} \end{pmatrix}$ and $\widehat{\Sigma} = \begin{pmatrix} \widehat{\sigma}_{11} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \widehat{\sigma}_{KK} \end{pmatrix}$ estimated by each GNS for $k = 1, ..., K$.

S2 Find the optimal parameters $(\boldsymbol{\beta},\ \Sigma)$ given matrices R and Λ in the below.

(c) Obtain $\widehat{\boldsymbol{\beta}} = \widehat{\boldsymbol{\beta}}(R, \Lambda, \widehat{\Sigma})$ by (13), and $\widehat{\Sigma} = \widehat{\Sigma}(R, \Lambda, \widehat{\boldsymbol{\beta}})$ by (15).

(d) Repeat (c) until convergence.

S3 Obtain the maximized profile given $(R,\ \Lambda)$ using the previous step.

$$\ell(R, \Lambda) = \ell(R, \Lambda, \widehat{\boldsymbol{\beta}}, \widehat{\Sigma}) = -\frac{Kn}{2}\log(2\pi e) + \log|I - R \otimes W| + \log|I - \Lambda \otimes W| - \frac{n}{2}\log|\widehat{\Sigma}(R, \Lambda)|$$

S4 Update (R, Λ) by applying a BFGS quasi-Newton method (Nocedal and Wright, 2006) to maximize $\ell(R, \Lambda)$. At iteration m, compute the search direction $d_m = -B_m \nabla \ell(R_m, \Lambda_m)$, where B_m is the inverse Hessian approximation, and perform a line search to obtain $(R_{m+1}, \Lambda_{m+1}) = (R_m, \Lambda_m) + \alpha_m d_m$, subject to the constraints $|\rho_{k\ell}| < 1$ and $|\lambda_{k\ell}| < 1$.

S5 Repeat S4 until convergence.

3.4. *Calculation of the determinant.* Although the spatial weight matrix W is fixed, the repeated evaluation of the determinant $\left|I_{Kn} - R \otimes W\right|$ is time-consuming when Kn becomes large. Using eigenvalues ω_i $(i = 1, ..., n)$ of W, we have the determinant can be expressed by traces r_j as

$$|I_{Kn} - R \otimes W| = \prod_{i=1}^{n} \left\{ 1 - \sum_{k=1}^{K} (-1)^j \ e_j(R)\ w_i^k \right\} \tag{18}$$

where $r_j = \text{trace}(R^j)$ for $j = 1, , , , K$. The $e_j(R)$ $(j = 1, ..., 7)$ are given by in the below.

$$e_1 = r_1, \quad e_2 = \frac{r_1^2 - r2}{2} \tag{19}$$

$$e_3 = \frac{r_1^3 - 3r_1 r_2 + 2r_3}{6}, \quad e_4 = \frac{r_1^4 - 6r_1^2 r_2 + 3r_2^2 + 8r_1 r_3 - 6r_4}{24} \tag{20}$$

$$e_5 = \frac{r_1^5 - 10r_1^3 r_2 + 15r_1 r_2^2 + 20r_1^2 r_3 - 20r_2 r_3 - 30r_1 r_4 + 24r_5}{120} \tag{21}$$

$$\begin{aligned} e_6 = \frac{1}{720}\big(&r_1^6 - 15r_1^4 r_2 + 45r_1^2 r_2^2 - 15r_2^3 + 40r_1^3 r_3 - 120r_1 r_2 r_3 + 40r_3^2 - 90r_1^2 r_4 \\ &+ 90r_2 r_4 + 144r_1 r_5 - 120r_6\big) \end{aligned} \tag{22}$$

$$\begin{aligned} e_7 = \frac{1}{5040}\big(&r_1^7 - 21r_1^5 r_2 + 105r_1^3 r_2^2 - 105r_1 r_2^3 + 70r_1^4 r_3 - 420r_1^2 r_2 r_3 + 280r_2^2 r_3 + 210r_1 r_3^2 \\ &- 210r_1^3 r_4 + 630r_1 r_2 r_4 - 420r_3 r_4 + 504r_1^2 r_5 - 504r_2 r_5 - 840r_1 r_6 + 720r_7\big) \end{aligned} \tag{23}$$

The determinant $\left|I_{Kn} - \Lambda \otimes W\right|$ is similarly obtained.

3.5. *Induced models from Multivariate Spatio-Temporal Regression Models.* The MGNS model proposed by Eqn (8) can represent a wide class of linear regression models by imposing suitable restrictions on the parameter matrices, thereby nesting both the MSAR and MSE models as special cases. These models are estimated similarly to Subsection 4.3.

Table 1 summarizes 11 regression specifications by assigning a four-character ID to each model. The four characters correspond, in order, to R (spatial autoregressive dependence), Λ (spatial dependence in the error term), A (temporal autoregressive dependence), and Σ (cross-equation error covariance), where each component is coded as a full $K \times K$ matrix (1), a $K \times K$ diagonal matrix (d), or the zero matrix (0). For example, the model ID 0011 represents a VARX with $R = \Lambda = \mathbf{0}$, while A and Σ are $K \times K$ parameter matrices; in contrast, dddd denotes the model in which R, Λ, A, and Σ are all diagonal, implying that the K response variables follow independent SDE models. The second, third and fourth rows of Table 1 give constraints imposed on MGNS model, the number of parameters and descriptions of each model.

TABLE 1
Model IDs of K-variate GNR models

ID	Constraints	No. of parameters	Model name
0011	$R = \Lambda = \mathbf{O}$	$p + K^2 + K(K+1)/2$	VARX
1011	$\Lambda = \mathbf{O}$	$p + 2K^2 + K(K+1)/2$	MSAR
0111	$R = \mathbf{O}$	$p + 2K^2 + K(K+1)/2$	MSEM
1111	—	$p + 3K^2 + K(K+1)/2$	MGNS
1001	$\Lambda = A = \mathbf{O}$	$p + K^2 + K(K+1)/2$	MSAR without time-AR
0101	$R = A = \mathbf{O}$	$p + K^2 + K(K+1)/2$	MSEM without time-AR
1101	$A = \mathbf{O}$	$p + 2K^2 + K(K+1)/2$	MGNS without time-AR
000d	$R = \Lambda = A = \mathbf{O}$, Σ:diagonal	$p + K$	Respective regression models
d0dd	R:diagonal, $\Lambda = \mathbf{O}$, A andΣ:diagonal	$p + 2K + K$	Respective SAR
0ddd	$R = \mathbf{O}$, $\Lambda, A,$ and Σ:diagonal	$p + 2K + K$	Respective SEM
dddd	$R, \Lambda, A,$ and Σ:diagonal	$p + 3K + K$	Respective GNS

Note: p denotes a number of regression coefficients excluding time-AR terms.

3.6. *Model evaluation and explanatory-variable selection.* **Penalized information criteria**: Let $\ell(\widehat{\boldsymbol{\theta}})$ denote the maximized log-likelihood for the MGNS model with model ID 1111, where $\boldsymbol{\theta} = (\boldsymbol{\theta}_1', \boldsymbol{\theta}_2')'$ with

$$\boldsymbol{\theta}_1 = (\rho_{11}, \ldots, \rho_{KK}, \lambda_{11}, \ldots, \lambda_{KK})' : 2K^2 \times 1, \;\; \boldsymbol{\theta}_2 = (\boldsymbol{\beta}', \sigma_{11}, \ldots, \sigma_{KK})' : (p + K(K+1)/2) \times 1$$

Here, $\boldsymbol{\theta}_1$ denotes the spatial dependence parameters and $\boldsymbol{\theta}_2$ collects the regression and covariance parameters.

Let $\mathcal{I}_{11}(\widehat{\boldsymbol{\theta}})$ denote the block of the Hessian matrix corresponding to $\boldsymbol{\theta}_1$. To account for the regularization imposed on spatial parameters, we define the effective degrees of freedom as

$$d_{\text{eff}} = \operatorname{trace}\left[\mathcal{I}_{11}(\widehat{\boldsymbol{\theta}}) \left\{\mathcal{I}_{11}(\widehat{\boldsymbol{\theta}}) + \gamma I\right\}^{-1}\right] + p + \frac{K(K+1)}{2}, \quad (\gamma \geq 0). \tag{24}$$

The first term represents the contribution of the penalized spatial parameters, while the second term corresponds to the unpenalized regression and covariance parameters.

This additive decomposition reflects the fact that only spatial parameters are subject to shrinkage, whereas the remaining parameters are estimated without regularization and therefore contribute their full degrees of freedom. The trace term can be interpreted as the effective dimension of the spatial parameter space (see Hastie et al. (2009)).

Using this definition, the penalized information criteria are given by

$$\text{pAIC} = -2\,\ell(\widehat{\boldsymbol{\theta}}) + 2\,d_{\text{eff}}, \quad \text{pBIC} = -2\,\ell(\widehat{\boldsymbol{\theta}}) + \log(Kn)\,d_{\text{eff}}, \tag{25}$$

where Kn denotes the effective sample size. When $\gamma = 0$, the trace term reduces to $2K^2$, and d_{eff} coincides with the nominal parameter count. In the numerical studies below, we use penalized AIC and penalized BIC to compare candidate specifications.

The use of information criteria in spatial econometric models is well established (see, e.g., Anselin (1988)), and the present formulation extends these criteria by incorporating selective regularization effects through the effective degrees of freedom.

Pseudo coefficients of determination: Because the standard variance decomposition does not generally apply to spatial autoregressive specifications; see, for example, LeSage and Pace (2009) and Elhorst (2014), we report a averaged pseudo R^2 defined by

$$\bar{R}^2_{\text{pseudo}} = \sum_{k=1}^{K} R^2_{k,\text{pseudo}}/K. \tag{26}$$

where $R^2_{k,\text{pseudo}}$ are pseudo R^2 for k-th responses defined by $\mathrm{corr}(\boldsymbol{y}_k,\ \widehat{\boldsymbol{y}}_k)^2$. This statistic is used only as a descriptive measure of fit and should not be interpreted as an exact proportion of explained variation.

4. Model Identification of MGNR models.

In the previous section, MGNR model was proposed by Eqn (8). The parameters $\rho_{k,\ell}$ and $\lambda_{k,\ell}$ in MGNR both describe spatial relationships. Therefore, it is necessary to verify the identifiability of the $2K^2$ parameters. In this chapter, we establish the identifiability of MGNR using the generalized method of moments (GMM), see e.g. Lee (2010). Since the time index t is not required for the proof of identifiability, the subscript t is omitted in this section.

4.1. *Instrumental matrix and moment function in GMM.*

We define the following notation.

(27)

$$\boldsymbol{u}_k = W\boldsymbol{y}_k : n\times 1,\ \boldsymbol{v}_k = W^2\boldsymbol{y}_k : n\times 1,\ U_k = WX_k : p_k\times p_k,\ V_k = W^2X_k : p_k\times p_k$$

Then, let $H : Kn \times 3p$ be the instrumental matrix and $\boldsymbol{\tau}$ be a parameter vector defined by

$$H = H_{11}\oplus\cdots\oplus H_{KK} \ \text{ with } \ H_{kk} = \big[\,X_k\,;\,Y_k\,;\,Z_k\,\big] : n\times 3p_k \ \ (k=1,...,K) \tag{28}$$

$$\boldsymbol{\tau} = \big(\,\boldsymbol{\beta}',\ \rho_{11},\rho_{12},...,\rho_{KK},\ \lambda_{11},\lambda_{12},...,\lambda_{KK}\,\big)' : (p+2K^2)\times 1 \tag{29}$$

where "$\oplus$" denotes the direct sum of matrices. Then, the moment function $g(\boldsymbol{\tau})$ used for GMM estimation is defined as

$$g(\boldsymbol{\tau}) = \frac{1}{Kn}E\left[\,H'\big(I_{Kn}-\Lambda\otimes W\big)\left\{\boldsymbol{y}-(R\otimes W)\boldsymbol{y}-X\,\boldsymbol{\beta}\,\right\}\right] \tag{30}$$

4.2. *Sufficient conditions of identifiability of MGNR models.*

Now, we can obtain the identifiability theorem of MGNR.

THEOREM 1. *Suppose that $\boldsymbol{y} : Kn\times 1$ follow MGNR in MSTR framework defined by Eqn (8) with a non-singular matrix $I-\Lambda\otimes W$, where the error vector $\boldsymbol{\epsilon} : Kn\times 1$ follow a continuous distribution meeting $E\left[\,\boldsymbol{\epsilon}\,\middle|\,X\,\right] = \mathbf{0}_{Kn}$ and Cov$(\boldsymbol{\epsilon}) = \Sigma\otimes I_n$. Let H and H_k $(k = 1,...,K)$ be instrumental variables defined by Eqn (26). Then, the parameter vector $\boldsymbol{\tau}$ in Eqn (27) is identifiable if the following two sufficient conditions hold.*

SC1 $\quad rank\big(H'_{kk}\,X_k\big) = \dim(\boldsymbol{\beta}_k) = p_k$ *for all* $k = 1, ..., K.$
SC2 $\quad rank\big(H\big) = rank\big(H_{11} \oplus \cdots \oplus H_{KK}\ \big) \geqq\ 2K^2$

MGNR identifiability requires finite second moments and exogeneity; normality is not needed. The proof of the theorem for general K can be obtained by directly extending the proof for the binary case ($K=2$). In the next subsection, we provide the proof for the binary case.

4.3. *Proof of Theorem in Bivariate GNR case.* We will prove the case $K=2$. Let the spatial parameter vectors as follows.

$$\boldsymbol{\rho} = \big(\ \rho_{11}, \rho_{12}, \rho_{21}, \rho_{22}\ \big)' : 4 \times 1 \ \text{ and } \ \boldsymbol{\lambda} = \big(\ \lambda_{11}, \lambda_{12}, \lambda_{21}, \lambda_{22}\ \big)' : 4 \times 1 \tag{31}$$

Then, partial derivatives of the moment function (27) by $\boldsymbol{\tau}' = (\boldsymbol{\beta}',\ \boldsymbol{\rho},\ \boldsymbol{\lambda}')$ are given by

$$\partial g(\boldsymbol{\tau}) \,/\, \partial \boldsymbol{\beta}' = -\frac{1}{2n}\, E\left[\ H' \left(I_{2n} - \Lambda \otimes W\right) X\ \right] \tag{32}$$

$$\partial g(\boldsymbol{\tau}) \,/\, \partial \boldsymbol{\rho}' = -\frac{1}{2n}\, E\left[H' \left(I - \Lambda \otimes W\right) \begin{pmatrix} W\boldsymbol{y}_1 & W\boldsymbol{y}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & W\boldsymbol{y}_1 & W\boldsymbol{y}_2 \end{pmatrix} \right] \tag{33}$$

$$\partial g(\boldsymbol{\tau}) \,/\, \partial \boldsymbol{\lambda}' = -\frac{1}{2n}\, E\left[\ H' \begin{pmatrix} W\boldsymbol{\xi}_1 & W\boldsymbol{\xi}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & W\boldsymbol{\xi}_1 & W\boldsymbol{\xi}_2 \end{pmatrix} \right] \tag{34}$$

where $\boldsymbol{\xi}_1$ and $\boldsymbol{\xi}_2$ are residual vectors defined by Eqn (6).

We prove Theorem 1 in Subsection 5.2 using a Jacobian–rank argument by the following three steps.

Step 1 (Jacobian). Let $g(\boldsymbol{\tau})$ denote the (population) moment function defined by Eqn (27). Then, the Jacobian matrix defined by $D(\boldsymbol{\tau}) := \dfrac{\partial g(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}'}$ is calculated whose entries are given in Eqns (30) – (32).

Step 2 (Rank condition). Local identifiability of $\boldsymbol{\tau}$ at $\boldsymbol{\tau}_0$ follows if the Jacobian has full column rank at $\boldsymbol{\tau}_0$, i.e., rank $\big(D(\boldsymbol{\tau}_0)\big) = \dim(\boldsymbol{\tau}) = p+8$. Equivalently, the mapping $\boldsymbol{\tau} \mapsto g(\boldsymbol{\tau})$ is locally one-to-one in a neighborhood of $\boldsymbol{\tau}_0$. Note here that the non-singular matrix $(I - \Lambda \otimes W)$ holds rank$(H'(I-\Lambda\otimes W)) = $ rank(H'). Therefore, by the sufficient conditions SC1 and SC2 in the statements of Therem 1, we have rank $\big(D(\boldsymbol{\tau}_0)\big) = \dim(\boldsymbol{\tau}) = p+8$.

Step 3 (Generic identification). The rank condition above holds *generically* if there exists at least one parameter value for which rank$(D(\boldsymbol{\tau})) = \dim(\boldsymbol{\tau})$, and the set of parameter values at which the rank drops is contained in a proper algebraic variety (hence has Lebesgue measure zero). Therefore, MGNR is generically identifiable: for almost all admissible parameter values $\boldsymbol{\tau}$, the parameter vector is locally identified by the moment conditions.

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Appendix A. Penalized partial likelihood and effective degrees of freedom.

A.1 Penalized partial likelihood formulation. Let the parameter vector be partitioned as $\boldsymbol{\theta} = (\boldsymbol{\theta}_1', \boldsymbol{\theta}_2')'$, where $\boldsymbol{\theta}_1$ denotes spatial dependence parameters and $\boldsymbol{\theta}_2$ collects the remaining regression parameters.

We define the profile log-likelihood for $\boldsymbol{\theta}_1$ as

$$\ell_c(\boldsymbol{\theta}_1) = \ell(\boldsymbol{\theta}_1, \widehat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1)), \tag{35}$$

where $\widehat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1)$ maximizes $\ell(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$ with respect to $\boldsymbol{\theta}_2$.

We consider the penalized partial log-likelihood

$$\ell_{cp}(\boldsymbol{\theta}_1) = \ell_c(\boldsymbol{\theta}_1) - \frac{\gamma}{2}\boldsymbol{\theta}_1'\boldsymbol{\theta}_1. \tag{36}$$

Let $\widehat{\boldsymbol{\theta}}_1$ denote the maximizer of $\ell_{cp}(\boldsymbol{\theta}_1)$. The first-order condition is given by

$$\frac{\partial \ell_c(\boldsymbol{\theta}_1)}{\partial \boldsymbol{\theta}_1} - \gamma \boldsymbol{\theta}_1 = 0. \tag{37}$$

A.2 Hessian representation. Let

$$\mathcal{I}_c(\boldsymbol{\theta}_1) = -\frac{\partial^2 \ell_c(\boldsymbol{\theta}_1)}{\partial \boldsymbol{\theta}_1 \partial \boldsymbol{\theta}_1'} \tag{38}$$

denote the Hessian of the profile log-likelihood.

A first-order Taylor expansion around the true parameter $\boldsymbol{\theta}_1^0$ yields

$$0 = \frac{\partial \ell_c(\boldsymbol{\theta}_1^0)}{\partial \boldsymbol{\theta}_1} - \mathcal{I}_c(\boldsymbol{\theta}_1^0)(\widehat{\boldsymbol{\theta}}_1 - \boldsymbol{\theta}_1^0) - \gamma \widehat{\boldsymbol{\theta}}_1. \tag{39}$$

Rearranging terms gives

$$\widehat{\boldsymbol{\theta}}_1 = (\mathcal{I}_c + \gamma I)^{-1}\mathcal{I}_c \boldsymbol{\theta}_1^0 + (\mathcal{I}_c + \gamma I)^{-1}\frac{\partial \ell_c(\boldsymbol{\theta}_1^0)}{\partial \boldsymbol{\theta}_1}. \tag{40}$$

A.3 Effective degrees of freedom. The estimator can be written in a linearized form as

$$\widehat{\boldsymbol{\theta}}_1 = S\,\boldsymbol{\theta}_1^0 + \text{noise}, \quad S = (\mathcal{I}_c + \gamma I)^{-1}\mathcal{I}_c. \tag{41}$$

The matrix S plays the role of a smoothing (influence) matrix. Following the general theory of effective degrees of freedom in regularized estimation, we define

$$d_{\text{eff}} = \text{trace}(S) = \text{trace}\left\{(\mathcal{I}_c + \gamma I)^{-1}\mathcal{I}_c\right\}. \tag{42}$$

Using the cyclic property of the trace, this can be equivalently written as

$$d_{\text{eff}} = \text{trace}\left\{\mathcal{I}_c(\mathcal{I}_c + \gamma I)^{-1}\right\}. \tag{43}$$

A.4 Special cases. When $\gamma = 0$, we have $S = I$, and thus

$$d_{\text{eff}} = \dim(\boldsymbol{\theta}_1), \tag{44}$$

which coincides with the usual parameter count.

For $\gamma > 0$, the effective degrees of freedom are reduced, reflecting the shrinkage imposed on spatial parameters.

Q and A from/to Reviewers. Q1: The definition of effective degrees of freedom does not seem consistent with the estimation procedure.

A1: We thank the reviewer for this important comment. In the revised manuscript, we have clarified that the effective degrees of freedom is defined consistently with the penalized partial likelihood framework. Specifically, the Hessian of the profile log-likelihood is used in place of the full likelihood Hessian. A detailed derivation has been added in Appendix A.

Q2: Why is the trace formula used for the degrees of freedom?

A2: The trace-based definition follows the standard notion of effective degrees of freedom in regularized estimation, where the trace of the influence (or smoothing) matrix measures model complexity. This interpretation is now explicitly stated in Appendix A.3.

Q3: Is the proposed criterion theoretically justified?

A3: The proposed formulation is grounded in the general theory of penalized likelihood and profile likelihood inference. The resulting degrees of freedom coincide with the trace of the influence matrix derived from the linearization of the estimator, which is a standard justification in the literature. We have added a formal derivation in Appendix A.

Q4: What is new compared to existing penalized likelihood methods?

A4: The novelty lies not in the penalized likelihood itself, but in its structured application to spatio-temporal models, where only spatial dependence parameters are regularized while regression coefficients are profiled out.

Q5: Why partial likelihood?

A5: The partial likelihood formulation allows us to separate structural spatial dependence from regression effects, leading to more stable estimation and interpretable regularization.

Q6: Why can the degrees of freedom be decomposed additively?

A6: This follows from the fact that only spatial parameters are subject to regularization, while regression parameters are estimated without shrinkage and therefore contribute their full degrees of freedom.